



IR-CONTROLLED WATER SUPPLY

■ SANDIP NAIR

This circuit can be used in homes or small restaurants for any type of water supply unit, such as toilet flush or washbasin tap.

The circuit can be divided into three sections: power supply, transmitter and receiver.

The power supply section is built around IC 7805 (IC1). The input supply for IC1 may be provided by solar panel (SP) output of 12V or a 9V battery. Voltage regulator 7805 provides constant 5V DC to operate the circuit.

The transmitter section is built around timer 555 (IC2), npn transistor BC548, infra-red LED1 and some discrete components. Timer NE555 is wired in astable multivibrator mode to generate pulses of 38kHz frequency by adjusting preset VR1. Transistor T1 is used to drive IR LED1, which transmits modulated IR signals up to

a few metres.

The receiver section is built around TSOP1738, timer 555 (IC3) and some discrete components. IR receiver TSOP1738 detects the IR signal transmitted by IR LED1. The IR LED and the TSOP1738 are oriented face to face such that the IR beam transmitted by the IR LED directly falls on the TSOP1738.

The output of the IR receiver module is given to trigger pin 2 of monostable IC3. When IC3 is triggered, its output goes high for about ten seconds, depending on the values of resistor R6 and capacitor C6. Relay RL1 remains energised as long as the monostable output is high. As a result, the solenoid valve gets required supply through normally-opened (N/O) contacts of relay RL1.

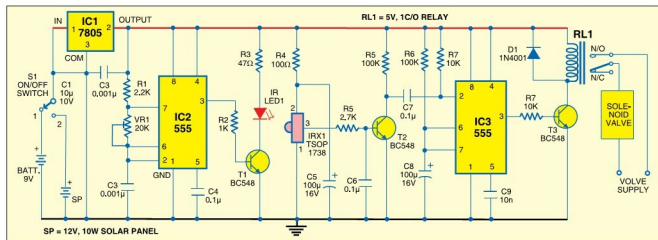
Depending on the type of the solenoid valve, the power supply requirement can be different. You can remove the relay if a 5V solenoid valve is used. In this case, you can connect the valve

between the collector of transistor T3 and positive side of 5V.

Thus when a person interrupts the IR beam falling from the IR transmitter (LED1) on IRX1 through his hand, the output of TSOP1738 goes high to trigger monostable IC3. The relay energises for about 10 seconds to actuate the solenoid valve to open the water supply. Water flows through the valve for 10 seconds and then the valve closes again.

If the hand is placed again during the 10-second period of water flow when the valve is opened, no action is taken by the monostable circuit. Rather, it continues to complete its function.

Assemble the circuit on a general-purpose PCB and enclose in a suitable cabinet. Fix the IR LED and IRX1 on the side walls of toilet flush/washbasin such that the transmitted IR beam falls directly on the receiver. ●



ELECTRIC GUITAR PREAMPLIFIER

■ T.K. HAREENDRAN

Here is the circuit of a guitar preamplifier that would accept any standard guitar

pickup. It is also versatile in that it has two signal outputs.

A typical example of using a pickup attached to a guitar headstock is shown in Fig. 1. The pickup device has

a transducer on one end and a jack on the other end. The jack can be plugged into a preamplifier circuit and then to

